

Plant Station R17 - Design Review

Controlled field	Value
Document number	PLANT-TR-001
Plant design revision	R17
Document issue	D02
Issue date	28 September 2026
Preceding document	PLANT-TR-001 R13 D01
Status	Issued for prototype review; physical acceptance open

Current mechanical configuration: R17 integrated body, R16 lid (R15 geometry), R13 fascia and planter. Electrical reference: frozen Uno UI v6 source and 27 September interface record. Fit, powered wet operation and unattended operation remain unaccepted.

R17 incorporates R15/R16 mechanical changes and the current Uno UI v6 interface. Previous issues are retained. Blank physical results remain open; no acceptance signature is inferred.

Revision control

Narrow parallel left-margin bars identify surviving changes against the verified matrix. Dates are documentary dates. R15/R16 source changes are incorporated without creating intervening manual issues.

Revision / class	Change and affected content
R5-R13 / source history	Earlier assembly and mechanical changes remain traceable through the preserved D01 matrix. Missing document editions are not invented.
R14 / preceding manual	AM R14 D01: Uno mounting and motor-disabled bench v2. TR's latest located same-family PDF is R13 D01; no TR R14 PDF is assumed.
R15 / 27 Sep 2026	Reinforced Uno bosses, thicker recessed-head lid and additional pump relief; source changes incorporated at R17, physical trials open.
R16 / 27 Sep 2026	Four relay bosses below Uno; source change incorporated at R17; actual underside and hole pitch unmeasured.
UI v6 / 27 Sep 2026	Three buttons and integrated lamps, enabled active-HIGH relay, 1-4 s request, soil display and 60 s backlight timeout. Firmware revision is independent of mechanical R17.
R17 / 28 Sep 2026	Mounted USB-C cut and pilots, adjacent diameter 10 DC extension hole; estimated dimensions authorised; coupon fit required.
D02 / 28 Sep 2026	Integrated current R17 document issue. Uses the established enhanced template, controls and cumulative revision bars. No R15/R16 manual issues created.

D02 / document control: new integrated R17 technical content in the established D01 A4 template. Pagination, new headings and editorial reflow alone receive no technical bar.

R5 is the earliest located assembly guide. R1-R4 and R9-R12 PDFs were not located in the previous inventory. R15/R16 are evidenced mechanical prototype changes, incorporated here without fictitious intermediate PDF/manual editions. Margin labels identify documentary changes, not physical test passes.

List of effective pages

All pages are effective at D02, dated 28 September 2026, for the R17 prototype. Section references provide navigation; technical change labels are controlled separately.

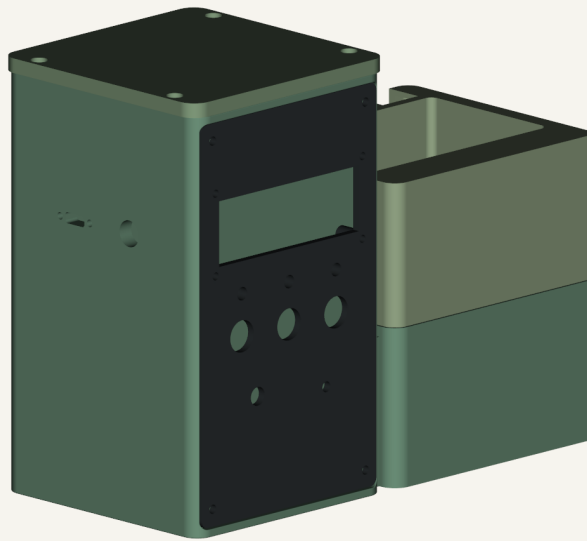
Page	Issue	Content
1	D02	Control pages
2	D02	Control pages
3	D02	Control pages
4	D02	Section 1
5	D02	Section 2
6	D02	Section 3
7	D02	Section 4
8	D02	Section 5
9	D02	Section 6
10	D02	Section 7
11	D02	Section 8

Section / page cross-reference

Section	Subject	Effective pages
1	Current state and incorporated changes	4
2	Known constraints and configuration contract	5
3	Mount, lid and pump options	6
4	USB-C and barrel panel decision	7
5	Electrical and firmware configuration impact	8
6	Evidence and digital verification limits	9
7	Risks, unknowns and closure actions	10
8	Acceptance recommendation and next actions	11

01 Current state and incorporated changes

R17 USB-C + DC PANEL assembly review



Provisional USB dimensions R16 relay / R15 mounts and lid retained

Actual assembled STL review; electronic hardware omitted.

R17

R17 incorporates R14 Uno registration, R15 splitting/lid/pump rectifications, R16 relay mounting and the R17 two-port panel. Current firmware scope is Uno UI v6. Digital geometry is available for prototype trials; physical fit and powered wet acceptance remain open.

R15

R14 was the reported printed baseline. Its Uno supports split during screw insertion. The user requested a thicker recessed-head roof and removal of the pump/tube obstruction; R15 records these changes. Screw dimensions, printed material and causal strength evidence are still absent.

This is the next located design-review document after TR R13 D01. R14-R16 geometry/evidence is incorporated without asserting missing same-family PDFs. R17 documents supersede their current assembly instructions, not the retained historical evidence.

02 Known constraints and configuration contract

			Interface	Current basis	Evidence status
R15 R13 R12	R15 R14 R15 R16 R17 R17 R12	R15	Body	207 x 100 x 158; integrated tower/sump	Confirmed export bounds
		Lid	99 x 100 x 5.5; top z163.5	Confirmed retained R15/R16 CAD	
		R14	Uno	Four asymmetric R14 centres; y76 seat	Source-verified; actual variant fit open
		R15	Bosses	8.5 tip / 10 root / 19 projection; 2.8 x 11 pilot	Prototype; physical screw fit open
		R16	Relay	50 x 26 PCB; 44.5 x 20.5 pitch	Seller/photo nominal; underside unknown
		R17	USB panel	9.8 x 4.2 R1.2, 2.8 pilots at 15.2 pitch	Authorised best guesses
		R17	DC panel	Diameter 10; centre 28 from USB	User hole size; nut/grip unknown
			Wet route	4 floor; separate tube/wire paths; sensor edge z60	Geometry checked; no fill limit

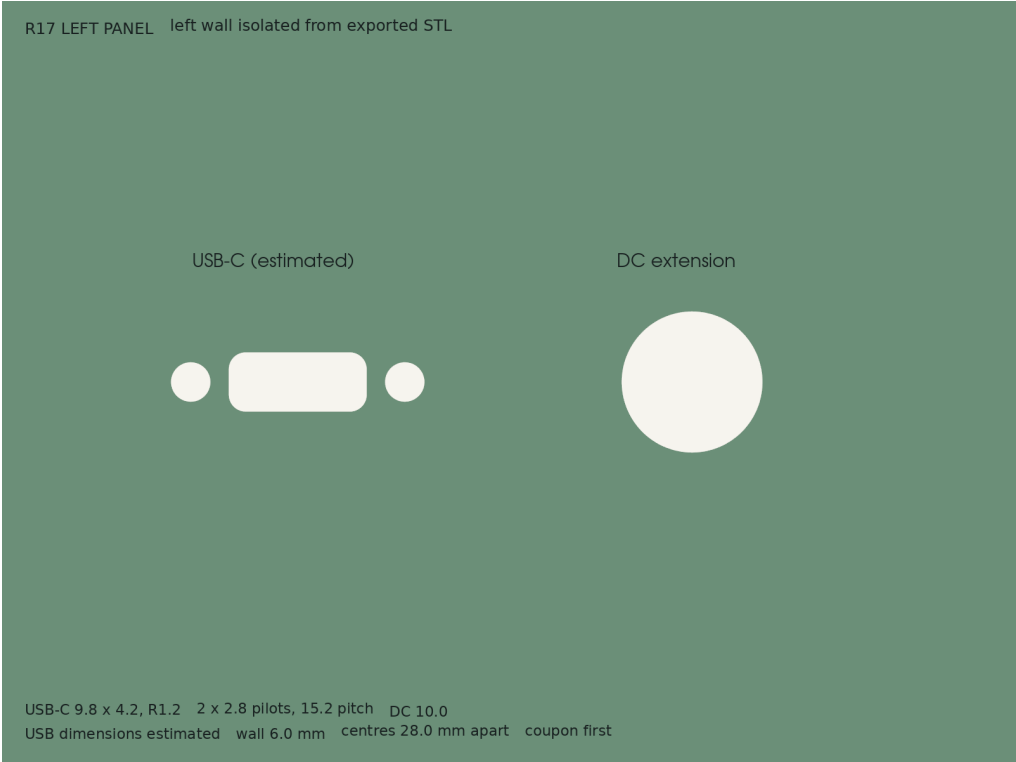
Preserve the wet/dry partition, planter serviceability, fascia interface and asymmetric board registration. Physical hardware measurements outrank the provisional proxy dimensions. No new firmware modification is included in R17 documentation.

03 Mount, lid and pump options

	Feature	Options and trade-off	Selected prototype / gate
R15	Uno mount	Thicker/tapered bosses retain four-screw assembly; inserts/captive nuts improve repeated service but need new fit/access data	R15 reinforced bosses retained. Coupon required; splitting cause is unproven
R15	Lid	Thicker pocketed lid gives flush heads; unchanged thin roof preserves old shape but misses request	R15 5.5 thick, 6.6 x 2.5 pockets; head/engagement trial
R15	Pump route	Rotate pump may avoid relief but restrict service; lower local relief removes reported ledge	R15 relief x106.5-132/y83-94/z4-52; actual attached tube trial
R16	Relay placement	Vertical below Uno consumes nominal height; side wall conflicts with access; rear-wall horizontal gives service space	R16 horizontal, 12 mm above saddle and 13 below Uno; terminal/wire fit open

Recommendation: retain the selected R15/R16 geometry for representative trials. Do not switch fastener architecture or widen pilots in the body without actual trial evidence and regenerated exports. Fit and retention are separate from digital solid validity.

04 USB-C and barrel panel decision



Left-wall mesh detail; both openings are at z108.5.

Option	Benefit	Trade-off
Integral wall cuts (selected)	Matches requested fixed panel; no extra printed part	Estimated USB fit and 6 mm DC grip must pass coupon
Replaceable connector insert	Easier later adjustment to another USB/DC component	Adds fastening, seams and another controlled part
Retain broad cable opening	Accepts direct original cable access	Does not implement requested fixed sockets

R17

The user requested the fixed USB/DC panel and explicitly authorised best guesses with PCB-sized holes. Integral wall cuts are selected for this prototype. USB shell cut 9.8 x 4.2 R1.2; pilots 2.8 through at 15.2 pitch; DC diameter 10. USB y63, DC y35, common z108.5.

R17

Estimated pilot-to-cut material is 1.3 mm. Image flange width 20.1 gives 12.95 mm to the DC cut; a guessed diameter-20 nut would leave 7.95 mm. These calculations depend on unmeasured component envelopes and do not establish fit or strength.

R17

CAUTION
Recommendation: coupon both actual connectors together, including mating plugs, nut grip, screw tips and solder service. Keep the electrical pin/power interface open until characterised; no USB-C rating or source behaviour follows from the cut.

05 Electrical and firmware configuration impact

R17	R14 disabled-motor bench v2 and spare third switch are historical. Current source enables active-HIGH D9 relay requests and uses D2/D3/D4 switches; D5/D6/D7 red/yellow/green lamps. A4/A5 serve LCD; A0/A1 serve probe/pot; D10 is unused.
R17	Dose requests are 1-4 s, knob sampled once. Yellow cycles pages/holds lamp test/cancels; STOP cancels and two-second hold clears session counters. Sensor is advisory; below 25% red blink, at/above green, raw ≤ 20 fault. Software STOP and sensor fault do not provide hardware isolation.
R17	UI v6 commands soil Status return at request end and I2C backlight off after 60 s idle, waking on buttons or ≥ 8 ADC pot-count change. Serial OFF/ON and LCD status 0 were recorded on 27 September; earlier communication failures and visual/post-dose checks remain open.

CAUTION

Relay module/pump rating, current, supply behaviour and reset default-OFF are unmeasured. USB-C pins/configuration and barrel polarity/grip are unresolved. CAD provision does not accept a new power system. Keep a physical pump disconnect.

06 Evidence and digital verification limits

	Evidence	Recorded result	Limit
R17	R15 independent checker	Six final meshes/sections passed	Fastener, pump and leak physical gates open
R17	R16 independent checker	Three single-component meshes and relay sections passed	Nominal board envelope only; underside unmeasured
R17	R17 STEP/readback	Both exports reopened as one valid solid	Builder process still exits 1 on shutdown
R17	R17 independent STL	Two closed consistently wound meshes; all edges two-face; no degenerate faces	Full mesh self-intersection not checked
R17	R17 sections	Six wall sections: USB 9.800 x 4.200, DC approx 9.998 centre section, pilots approx 2.800	Tessellated mesh dimensions; not printed-hole readings
R17	R17 change scope	0 added/removed mm3 outside left-wall mask	Does not establish physical insertion or strength
R17	UI v6 source and record	Current SHA-256 matches 27 September upload record	No new hardware run in this documentation task

Nominal module proxies have zero digital interference with the body/board proxies. Actual component fronts, solder, plug bodies and service loops are not represented accurately enough for physical acceptance. No slicing, physical print, continuous swept-lift or wet test is claimed.

07 Risks, unknowns and closure actions

ID / type	Condition and effect	Mitigation / closure
RI17-01 / Issue	USB cut/pitch and jack grip unmeasured; mismatch could require a body reprint	A17-01 coupon plus actual measurements
RI17-02 / Issue	Reported R14 boss splitting; R15 reinforcement physically untested	A17-02 screw/material/orientation trial; inspect root and tip
RI17-03 / Risk	Relay solder/terminals and fascia services may collide	A17-05/06 full assembled dry fit and removal trial
RI17-04 / Issue	LCD intermittent communication faults; screen may fail under load	A17-09 supply/I2C/suppression investigation and repeat observations
RI17-05 / Risk	Relay closure through boot/reset or welded contacts may leave pump running	A17-08 measured default-OFF; physical disconnect
RI17-06 / Issue	Fill/leak/material limits unknown; ingress into electronics possible	A17-11/12 material qualification and controlled leak/drain-back/splash evidence
RI17-07 / Issue	USB/barrel wiring and rating unresolved	A17-07 identified source diagram, continuity and load evidence
RI17-08 / Issue	CAD exit code 1 prevents a clean-build claim	A17-13 isolate shutdown cause or retain explicit exception

Risk ratings are qualitative; no likelihood/severity matrix was supplied. All owners are Unassigned, target dates Not set, and all entries remain Open. USB guesses, relay nominal dimensions, screw torque, printer/material, supply/load limits and acceptance tolerances remain Unknown or Assumed as identified.

08 Acceptance recommendation and next actions

1. Print and fit the R17 port coupon; in parallel complete the matching Uno, lid, pump and relay trials. Record measurements and photographs. 2. Correct failed dimensions and re-export only affected geometry. 3. Complete full-body slicing review and dry serviceability fit. 4. Close the electrical interface and relay default-OFF evidence before supervised powered testing. 5. Close leak/fill and dose-volume evidence before accepted supervised wet use.

PLANT-TS-001 contains testable A17-01 to A17-13 records. A recommendation is not an acceptance signature. Current disposition: prototype review; digital evidence recorded; physical/electrical acceptance open.

References are frozen in the package evidence register. Prior enhanced D01 scheme/matrix is retained alongside the extended R17 change matrix. No R15/R16 manual issues or missing early PDFs were created.